Reg. No. | | | | | | | | | | [MSCCH201]

SECOND SEMESTER UNIVERSITY EXAMINATION, JUNE 2016

M.Sc. (CHEMISTRY)

INORGANIC CHEMISTRY-II

Time : 03 Hours **Maximum Marks:80**

Instructions: **1.** Questions 1 to 6 carry 15 marks each, answer any four.

2. Question 7 is compulsory, carries 20 marks.

1. **a.** Give detailed account on coordination numbers and coordination geometries.

b. What is chelate effect? Explain with appropriate examples. **10+5**

2. **a.** Explain the factors affecting the crystal field splitting of octahedral complexes.

b. Account for the following,

i. $[Ni(CNS)_4]^{2-}$ is square-planar

ii. $[Co(NH_3)_6]^{2+}$ is octahedral. **10+5**

3. **a.** Define stereo center, chiral center and circular birefringence and give the differences between chiral and stereo centers.

b. Give a general graphical representation for UV, CD, ORD and Cotton curves, and explain the changes observed in each of them.

c. Define CD, ORD and MCD? Explain their principles. **5+5+5**

4. **a.** Determine the ground state term symbol and draw energy level diagram for $[V(H_2O)_6]^{3+}$, and possible electronic transitions.

b. Explain Laporte orbital and spin selection rules.

c. Draw combined Orgel level diagram for d^2, d^3, d^7 and d^8 cases. **5+5+5**

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5. **a.** Write a short note on spin cross over complexes.

b. List the differences between ferromagnetism and ferrimagnetism.

c. Describe the Faraday's method to determine the magnetic susceptibility of coordination compounds. **5+5+5**

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6. a. Write a note on isolobal analogy of organometallic compounds.

b. Calculate the number of bonds present between the metal centers in the following,

i. $[CpCr(CO)_2]_2$

ii. $[Fe_3(CO)_{12}]$.

c. Calculate the total number of valance electrons present in the following,

i. $[Co(PPh_3)_2(CO)(allyl)]^{1+}$

ii. $[Ni(allyl)(H_2O)_2]^{1+}$.

7. **Answer any TEN of the following.** **10x2=20**

a. Define ambidentate ligand with an example.

b. Explain the term denticity with an example.

c. $[Zn(en)_3]^{2+}$ is white in colour. Justify.

d. $[Co(H_2O)_6]^{2+}$ is paramagnetic in nature. Give reason.

e. Explain umbrella structure of tertiary phosphines.

f. Define molar ellipticity and write its equation.

g. Explain spin selection rule.

h. Explain the colour of $[Co(NH_3)_5Cl]^{3+}$ and $[MnBr_4]^{2-}$.

i. Write the equation for the determination of spin-orbit coupled magnetic moment and explain its terminologies.

j. Calculate the spin-only magnetic moment value for $[Cu(NH_3)_4]SO_4$.

k. Calculate the total number of valance electrons present in $[Ni(CH_3)_2(PPh_3)_2]$.

l. Write the coordination number and oxidation state of the metal center for $[Co(CO)_2(allyl)(PPh_3)Cl]^{+}$.